

Comparative Study of Classical NLP and Pretrained Language Models for Tunisian Dialect Processing

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Abstract

Recent transformer-based language models rely on subword tokenization and contextual embeddings, raising questions about the continued effectiveness of traditional preprocessing techniques such as stemming. This issue remains underexplored for low-resource Arabic dialects, particularly Tunisian Dialect (TD). In this paper, we investigate the impact of light and aggressive stemming on two main Tunisian NLP tasks: sentiment analysis and domain classification. We compare classical machine learning methods with transformer-based Arabic language models, including TunBERT. To support this study, we release an open Tunisian stemming resource containing 7,730 words annotated with both light and aggressive stems. Our experiments provide a comprehensive analysis of TD processing and show that the effectiveness of stemming depends on both the task and the model type.

1 Introduction

The emergence of embedding-based models and pretrained language models has significantly reduced the reliance on traditional machine learning pipelines and manual preprocessing techniques. These models, particularly transformer-based architectures, achieve strong performance across many Natural Language Processing (NLP) tasks by learning contextualized representations from large-scale corpora (Alkaabi et al., 2025).

However, low-resource languages and dialects such as Tunisian Dialect (TD) remain particularly challenging for NLP due to their rich morphology, lack of standardization, and high lexical variability. Moreover, they suffer from a severe scarcity of annotated and open-source resources (Besdouri et al., 2021; Ibn Nasr et al., 2025; Besdouri et al., 2024). Traditionally, Arabic NLP pipelines addressed these challenges by relying heavily on preprocessing techniques such as stemming and feature engineering to enhance the performance of

classical machine learning models (Habash, 2005), which has raised questions about the continued necessity and effectiveness of such traditional preprocessing approaches in the presence of recent language models (Alkaabi et al., 2025; Atwan et al., 2021b).

In this work, we investigate the impact of stemming on TD NLP by systematically evaluating light and aggressive stemming across classical machine learning and transformer-based models, using a newly released open stemming resource for TD. The study evaluates these strategies on sentiment analysis and domain classification tasks, providing both a reusable linguistic resource and an empirical assessment of stemming techniques.

The main contributions of this paper are as follows:

- We investigate the impact of stemming on TD NLP.
- We compare classical machine learning under preprocessing and transformer-based models.
- We evaluate these strategies on sentiment analysis and domain classification tasks.
- We release an open stemming resource containing light and aggressive stems for TD with 7,730 instances.

The remainder of this paper is organized as follows: Section 2 presents the related work; Section 3 describes the resources and methodology; Section 4 presents the results; Section 5 discusses the findings; and Section 6 concludes the paper and outlines future directions.

2 Related Work

Arabic is a morphologically rich language, making preprocessing techniques important for NLP tasks (Habash, 2010). Stemming techniques are

commonly classified into light stemming, which removes affixes, and aggressive stemming, which attempts to recover roots (Mustafa et al., 2017). Various stemmers, including Khoja, ISRI, and Light10, have been widely used to reduce lexical sparsity and improve text classification and information retrieval performance (Mustafa et al., 2017; Alayba, 2025; Naili et al., 2019; Atwan et al., 2021a,b). However, aggressive stemming may introduce ambiguity and reduce performance (Atwan et al., 2021b; Naili et al., 2019).

Transformer-based models such as AraBERT (Antoun et al., 2020), MARBERT (Abdul-Mageed et al., 2021), CAMeLBERT (Inoue et al., 2021), and TunBERT (Messaoudi et al., 2022) have substantially advanced Arabic NLP. Their contextual representations and subword tokenization can capture morphological information without explicit normalization (Abdelali et al., 2022). As a result, the role of traditional preprocessing remains debated (Alkaabi et al., 2025). While stemming often benefits sparse representations such as TF-IDF (Atwan et al., 2021a), excessive preprocessing may remove useful semantic information (Alayba, 2025), particularly in dialectal Arabic, where limited resources remains a major challenge.

3 Data and Methodology

This section presents the components of the proposed methodology, as illustrated in Figure 1, detailing the process from the NLP tasks to model evaluation.

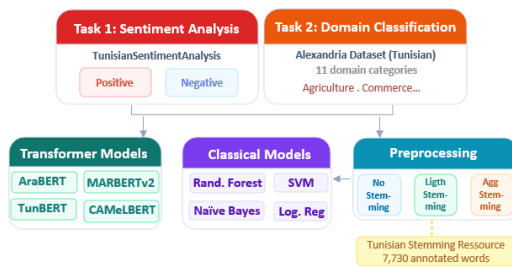


Figure 1: Workflow for TD Downstream NLP Tasks Evaluation

3.1 Resources

3.1.1 Tunisian Stemming Resource

To conduct our experiments, we release an open Tunisian stemming resource composed of 7,730 Tunisian words manually annotated with both light and aggressive stems. Each entry contains three fields: (i) the original Tunisian word, (ii) its light stem, and (iii) its aggressive stem. Table 1 presents illustrative examples from the corpus.

The corpus was manually constructed following a controlled annotation protocol involving two native Tunisian annotators and one adjudicator responsible for resolving disagreements. Annotators were selected based on their expertise in linguistic analysis, and all were graduates of the Faculty of Letters with a specialization in linguistics. Prior to annotation, a calibration meeting was held to establish common guidelines and ensure consistency.

The resource was developed in two stages: manual collection and data enrichment. In the manual collection stage, native-speaking annotators collected words from everyday Tunisian communication, covering common lexical and morphological patterns. In the data enrichment stage, additional lexical items were extracted from the Tunisian STT-TTS dataset¹ using the Camel Tools library for tokenization. Duplicates were removed, and inconsistent or non-standard entries were corrected or excluded.

To assess annotation reliability, we computed Cohen’s Kappa (κ) (Cohen, 1960), obtaining an agreement score of 93%, indicating high inter-annotator consistency. Remaining disagreements were resolved by the adjudicator.

An intrinsic evaluation was conducted on a held-out subset of 800 words from the annotated corpus. The hybrid stemmer achieved 92.8% accuracy for light stemming and 86.1% for aggressive stemming against the manually annotated reference stems. Further details on the evaluation protocol, disagreement resolution, results, and qualitative analysis of stemming transformations and errors are provided in Appendix A.

The dataset is publicly available under a CC-BY license at: <https://github.com/LatifaIbenNasr/Stemming>.

3.1.2 Downstream NLP Tasks

We conduct experiments on two Tunisian NLP tasks: sentiment analysis using the TunisianSentimentAnalysis dataset², which contains Tunisian social media texts annotated with two classes: positive and negative, and domain classification using the Tunisian subset of the Alexandria dataset (Mekki et al., 2026), which includes 11 domain categories (e.g., Agriculture / Farming, Commerce / Transactions) for each text.

¹https://huggingface.co/datasets/Arbi-Housseem/Tunisian_dataset_STT-TTS15s_filtred1.0

²<https://huggingface.co/datasets/hedhoud12/TunisianSentimentAnalysis>

Words'origin	Light Stem	Aggressive Stem
جايبينك / jAybynk / we brought you	جاين / jAyiy / coming	جاي / jay / come
الصحونة / alSaHwnp / the dishes	صحونة / SaHwnhp / dishes	صحن / SaHn / dish
الكرهيب / alkrAheb / the cars	كراهب / krAheb / cars	كرهبة / krhbp / car
البرودويات / AlbrwdwyAt / the products	برودويات / brwdwyAt / products	برودوي / brwdwY / product

Table 1: Examples of Stemming in the TD

3.2 Models

We compare two categories of NLP models: classical machine learning and transformer-based models.

3.2.1 Classical Machine Learning Models

We evaluate four classical models: Support Vector Machines (SVM), Logistic Regression (LR), Naive Bayes, and Random Forest. All models use CAMEL tokenization followed by TF-IDF feature extraction.

Hyperparameters are set as follows: SVM uses a linear kernel with $C = 1.0$; LR uses the *liblinear* solver with $C = 1.0$; Multinomial Naive Bayes uses $\alpha = 1.0$; and Random Forest uses 100 trees with a maximum depth of 20. The dataset is split into 70% training, 15% validation, and 15% testing sets.

3.2.2 Transformer-based Models

We evaluate four transformer models: AraBERT, MARBERTv2, CAMELBERT, and the Tunisian-specific TunBERT. All models are fine-tuned separately for each task.

The training setup is identical across models: maximum sequence length of 128 tokens, batch size of 16, learning rate of 2×10^{-5} , and 3-5 training epochs depending on convergence. We use the Adam optimizer with linear learning rate decay and warmup steps set to 10% of total training steps.

3.3 Preprocessing Configurations

To analyze the impact of TD stemming, we evaluate three configurations: no stemming, light stemming, and aggressive stemming.

For stemming, we adopt a hybrid approach that combines exact dictionary lookup using the proposed Tunisian stemming resource with fuzzy string matching. If a token is absent from the stem lexicon, the closest lexical entry is retrieved using the `fuzz.ratio` function from the `RapidFuzz` library, which computes a normalized Levenshtein-based similarity score. The stem of the closest entry is assigned only if the similarity score exceeds a

predefined threshold. The fuzzy matching threshold was tuned on the development set by evaluating values of 75, 80, 85, 90, and 95. A threshold of 85 was selected, as it provided the best trade-off between stem recovery and erroneous stem assignments.

The `fuzz.ratio` metric was selected to measure string similarity and accommodate minor orthographic differences observed in TD (Turki et al., 2024). Its normalized similarity score and efficient implementation make it suitable for the fuzzy matching step (Bachmann et al., 2025).

4 Results

For sentiment analysis, as shown in Tables 2 and 3, the results indicate that stemming does not improve the results. The best classical result is achieved by Naive Bayes without stemming, reaching an accuracy of 0.8737 and an F1-score of 0.8702. Transformer models obtain slightly higher scores, with AraBERTv02 achieving the best overall performance (0.8834 accuracy).

Model	Metric	No stemming	LightStem	AggStem
SVM	Acc	0.8707	0.8669	0.8674
	F1	0.8691	0.8654	0.8658
LR	Acc	0.8686	0.8657	0.8655
	F1	0.8644	0.8623	0.8622
Naive Bayes	Acc	0.8737	0.8707	0.8650
	F1	0.8702	0.8662	0.8605
Random Forest	Acc	0.8409	0.8369	0.8430
	F1	0.8347	0.8306	0.8370

Table 2: Performance of classical machine learning models under preprocessing strategies for sentiment analysis.

Model	Accuracy	F1 Score
MARBERTv2	0.8792	0.8790
AraBERTv02	0.8834	0.8833
CAMELBERT	0.8651	0.8653
TunBERT	0.6931	0.6871

Table 3: Performance of transformer-based models on the sentiment analysis task

Figure 2 summarizes the accuracy comparison across all evaluated models. It demonstrates that traditional approaches remain competitive for this task, with a modest performance for TunBERT.

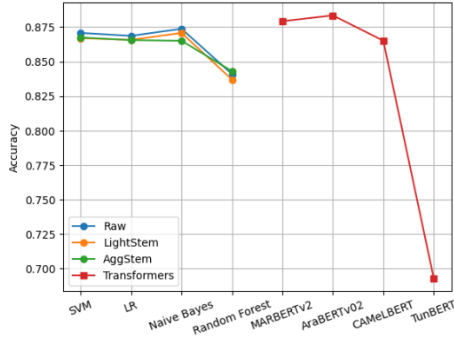


Figure 2: Accuracy comparison on the sentiment analysis task.

For domain classification, as illustrated in Tables 4 and 5, preprocessing has a stronger impact. Light stemming consistently improves performance, allowing SVM to achieve the best classical result (0.4788 accuracy). In contrast, aggressive stemming provides smaller gains and suggests that it may remove discriminative lexical information that helps distinguish between domains. Transformer models obtain lower scores, with AraBERTv02 and MARBERTv2 reaching the highest accuracy among them (0.3835).

Model	Metric	No stemming	LightStem	AggStem
SVM	Acc	0.4087	0.4788	0.4498
	F1	0.3983	0.4561	0.4304
LR	Acc	0.3639	0.4050	0.4182
	F1	0.3189	0.3659	0.3668
Naive Bayes	Acc	0.3296	0.3654	0.3865
	F1	0.2813	0.3136	0.3297
Random Forest	Acc	0.2926	0.3232	0.3073
	F1	0.2740	0.3112	0.2758

Table 4: Results of classical machine learning models on the domain classification task

Model	Accuracy	F1 Score
MARBERTv2	0.3835	0.3338
AraBERTv02	0.3835	0.3413
CAMELBERT	0.3383	0.2730
TunBERT	0.1053	0.0201

Table 5: Performance of transformer-based models on the domain classification task

As shown in Figure 3, the best-performing classical configuration outperforms all transformer-based models on this task. Although TunBERT was pretrained on Tunisian dialect data, it still achieves the lowest performance among the evaluated transformer models.

Overall, the results highlight the task-dependent effect of stemming. While sentiment analysis benefits from preserving original word forms, domain classification benefits from moderate normalization through LightStem. Transformer models per-

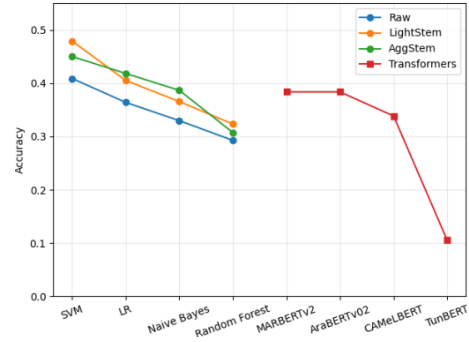


Figure 3: Accuracy comparison on the domain classification task.

form best on sentiment analysis, whereas classical machine learning, particularly SVM with LightStem, proves more effective for domain classification. Light stemming consistently outperforms aggressive stemming. Classical approaches remain competitive and, in some cases, outperform transformer models despite their lower computational requirements. These findings suggest that careful preprocessing and model selection remain crucial for Arabic downstream NLP tasks.

5 Discussion

Our results show that the effect of stemming is strongly task-dependent. For domain classification, light stemming improves the performance of SVM and other classical models, which is consistent with previous studies reporting benefits of stemming for Arabic text classification (Atwan et al., 2021a; Wahbeh et al., 2011; Ayedh et al., 2016). Domain classification relies heavily on domain-specific terminology, and light stemming can group morphological variants while reducing feature sparsity in TF-IDF representations without substantially altering lexical identity. In contrast, aggressive stemming may remove discriminative lexical information by reducing domain-specific terms to broader or semantically different roots. For example,

القمحية is preserved as قمحية by light stemming but reduced to قمح (wheat) by aggressive stemming, potentially introducing an agriculture-related interpretation. Similarly, الملاحة (navigation/shipping) is reduced to ملح (salt), while الجامعة is reduced to the generic root جمع. These cases illustrate how aggressive stemming can weaken domain-specific lexical cues, explaining why light stemming generally outperforms aggressive stemming in domain classification.

For sentiment analysis, stemming has a much more limited effect. Sentiment prediction depends more strongly on sentiment-bearing lexical forms and contextual expressions, for which morphological distinctions may carry useful information. Moreover, transformer models can capture substantial morphological variation through subword tokenization and contextual representations (Alkaabi et al., 2025). Thus, explicit stemming provides limited additional benefit for sentiment analysis and may sometimes remove useful lexical distinctions.

The performance of TunBERT provides further evidence that preprocessing and model vocabulary interact with the characteristics of the task. Despite being pretrained on TD data, TunBERT was consistently outperformed by AraBERT and MARBERT, which may be partly explained by the larger and more diverse pretraining resources available for these models (Abdul-Mageed et al., 2021; Inoue et al., 2021).

To investigate this performance gap, we analyzed tokenization on the Alexandria dataset. TunBERT showed a higher Out-of-Vocabulary (OOV) rate (3.8% vs. less than 0.5% for AraBERT and MARBERT) and greater subword fragmentation, with an average of 2.47 subwords per word, compared with 1.61 and 1.67, respectively. This suggests a vocabulary mismatch between TunBERT’s pretraining data and domain-specific expressions in the evaluation corpus.

Overall, these findings indicate that stemming should be adapted to both the downstream task and the model architecture. Light stemming provides a useful balance between morphological normalization and lexical preservation for classical lexical-feature-based models, whereas aggressive stemming can remove discriminative domain cues. Transformer-based models, in contrast, can handle much of the morphological variation through subword and contextual representations. Therefore, stemming should not be considered a universally beneficial preprocessing step, but rather a task- and model-dependent strategy.

6 Conclusion and Future Direction

This study evaluated the impact of stemming on TD NLP for sentiment analysis and domain classification. The results show that light stemming benefits classical machine learning models, particularly for domain classification, whereas transformer models are less affected. Nevertheless, classical ap-

proaches remain competitive when combined with suitable preprocessing.

Future work will extend the evaluation to larger and more diverse Arabic corpora and enrich the publicly available stemming corpus to support further research in Arabic NLP.

Limitations

This work is limited to two TD NLP tasks and a stemming resource of 7,730 annotated words. The resource may not fully represent the lexical and regional diversity of TD. Moreover, only a limited set of classical and transformer-based models was evaluated. Future work should investigate additional NLP tasks, larger datasets, and other Arabic dialects.

References

- Ahmed Abdelali, Nadir Durrani, Fahim Dalvi, and Hassan Sajjad. 2022. Post-hoc analysis of arabic transformer models. In *Proceedings of the Fifth BlackboxNLP Workshop on Analyzing and Interpreting Neural Networks for NLP*, pages 91–103.
- Muhammad Abdul-Mageed, AbdelRahim Elmadany, and 1 others. 2021. Arbert & marbert: Deep bidirectional transformers for arabic. In *Proceedings of the 59th annual meeting of the association for computational linguistics and the 11th international joint conference on natural language processing (volume 1: long papers)*, pages 7088–7105.
- Abdulaziz M Alayba. 2025. Arabic natural language processing (nlp): A comprehensive review of challenges, techniques, and emerging trends. *Computers*, 14(11):497.
- Hussein Ala’a Alkaabi, Ali kadhim Jasim, and Ali Darroudi. 2025. Arabic nlp: A survey of pre-processing and representation techniques. *Journal of Computer Science, Information Technology and Telecommunication Engineering*, 6(2):876–890.
- Wissam Antoun, Fady Baly, and Hazem Hajj. 2020. Arabert: Transformer-based model for arabic language understanding. In *Proceedings of the 4th workshop on open-source arabic corpora and processing tools, with a shared task on offensive language detection*, pages 9–15.
- Jaffar Atwan, Mohammad Wedyan, Qusay Bsoul, Ahmad Hamadeen, Ryan Alturki, and Mohammed Ikram. 2021a. The effect of using light stemming for arabic text classification. *International Journal of Advanced Computer Science and Applications*, 12(5):768–773.
- Jaffar Atwan, Mohammad Wedyan, Qusay Bsoul, Ahmad Hammadeen, and Ryan Alturki. 2021b. The use

- of stemming in the arabic text and its impact on the accuracy of classification. *Scientific Programming*, 2021(1):1367210.
- Abdullah Ayedh, Guanzheng Tan, Khaled Alwesabi, and Hamdi Rajeh. 2016. The effect of preprocessing on arabic document categorization. *Algorithms*, 9(2):27.
- Max Bachmann, Georgia Kokkinou, Henry Schreiner, Jeppe Fihl-Pearson, Michał Górny, Moshe Sherman, Cristian Le, Nicolas Renkamp, Kwuang Tang, Hugo Le Moine, and 1 others. 2025. rapidfuzz/rapidfuzz: Release 3.13.0. *Zenodo*.
- Fatma Zahra Besdouri, Asma Mekki, Inès Zribi, and Mariem Ellouze. 2021. Improvement of the cota-orthography system through language modeling. In *2021 IEEE/ACS 18th International Conference on Computer Systems and Applications (AICCSA)*, pages 1–7. IEEE.
- Fatma Zahra Besdouri, Inès Zribi, and Lamia Hadrich Belguith. 2024. Arabic automatic speech recognition: challenges and progress. *Speech Communication*, 163:103110.
- Jacob Cohen. 1960. A coefficient of agreement for nominal scales. *Educational and psychological measurement*, 20(1):37–46.
- Rehab Duwairi, Mohammad Al-Refai, and Natheer Khasawneh. 2007. Stemming versus light stemming as feature selection techniques for arabic text categorization. In *2007 Innovations in Information Technologies (IIT)*, pages 446–450. IEEE.
- Nizar Habash. 2005. Introduction to arabic natural language processing. In *Association for Computational Linguistics*, volume 30.
- Nizar Y Habash. 2010. *Introduction to Arabic natural language processing*. Morgan & Claypool Publishers.
- Latifa Ibn Nasr, Abir Masmoudi, and Lamia Hadrich Belguith. 2025. Emotion recognition from spontaneous tunisian dialect speech. *ACM Transactions on Asian and Low-Resource Language Information Processing*, 24(2):1–16.
- Go Inoue, Bashar Alhafni, Nurpeiis Baimukan, Houda Bouamor, and Nizar Habash. 2021. The interplay of variant, size, and task type in arabic pre-trained language models. In *Proceedings of the sixth Arabic natural language processing workshop*, pages 92–104.
- Abdellah El Mekki, Samar M Magdy, Houdaifa Atou, Ruwa AbuHweidi, Baraah Qawasmeh, Omer Nacar, Thikra Al-hibiri, Razan Saadie, Hamzah Alsayadi, Nadia Ghezaiel Hammouda, and 1 others. 2026. Alexandria: A multi-domain dialectal arabic machine translation dataset for culturally inclusive and linguistically diverse llms. *arXiv preprint arXiv:2601.13099*.
- Abir Messaoudi, Ahmed Cheikhrouhou, Hatem Hadad, Nourchene Ferchichi, Moez BenHajhmida, Abir Korched, Malek Naski, Faten Ghriss, and Amine Kerkeni. 2022. Tunbert: Pretrained contextualized text representation for tunisian dialect. In *International Conference on Intelligent Systems and Pattern Recognition*, pages 278–290. Springer.
- Mohammad Mustafa, Afag Salah Eldeen, Sulieman Bani-Ahmad, Abdelrahman Osman Elfaki, and 1 others. 2017. A comparative survey on arabic stemming: approaches and challenges. *Intelligent Information Management*, 9(2):39–78.
- Marwa Naili, Anja Habacha Chaibi, and Henda Hajjami Ben Ghezala. 2019. Comparative study of arabic stemming algorithms for topic identification. *Procedia Computer Science*, 159:794–802.
- Houcemeddine Turki, Kawthar Ellouze, Hager Ben Amar, Mohamed Ali Hadj Taieb, Imed Adel, Mohamed Ben Aouicha, Pier Luigi Farri, and Abderrezak Ben-nour. 2024. Normalized orthography for tunisian arabic. In *International Conference on Language Processing and Knowledge Management*, pages 133–142. Springer.
- Abdullah Wahbeh, Mohammed Al-Kabi, Qasem Al-Radaideh, Emad Al-Shawakfa, and Izzat Alsmadi. 2011. The effect of stemming on arabic text classification: An empirical study. *International Journal of Information Retrieval Research (IJIRR)*, 1(3):54–70.

A Appendix

Further details on stemming variation, intrinsic evaluation, and additional analyses are provided below.

A.1 Stemming Variation and Annotator Disagreement

Most disagreements were due to sociolinguistic variation between urban and rural Tunisian dialects, particularly in aggressive stemming. For example, for the words [نمشيو / nemshyw / we go] and [كلات / klet / she eats], the urban annotator produced the aggressive stems [مشى / mshY / to go] and [كلي / kly / to eat], whereas the rural annotator proposed [مشي / mshy / to go] and [كلي / kly / to eat]. This variation concerns the final vowel realization [ى / Y] versus [ي / y]. To ensure consistency and compatibility with Modern Standard Arabic (MSA) orthography, the adjudicator standardized these forms using the final [ى / Y] representation.

To investigate the behavior of stemming forms as illustrated in Figure A.1, we computed the Levenshtein distance between each original word and its stem for both light and aggressive stemmers.

On average, the aggressive stemmer alters words more extensively (mean distance 2.79) than the light stemmer (mean distance 1.64), and exhibits higher variability (standard deviation 1.34 versus 0.69), indicating that some words undergo substantial transformation.

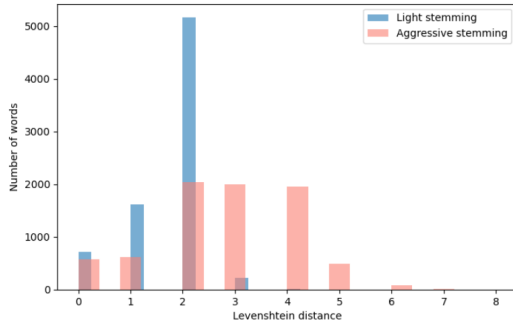


Figure A.1: Word Changes By Stemming Type

A correlation of 0.66 between the distances of the two stemmers suggests a moderate positive relationship: although both stemmers often target the same types of words, the aggressive stemmer applies more radical changes in certain cases. Examination of the most modified words shows that the aggressive stemmer sometimes produces stems with little semantic resemblance to the original, whereas the light stemmer generally preserves the lexical root.

These observations illustrate the trade-off between lexical simplification and semantic preservation, emphasizing that the choice of stemming strategy can significantly influence word forms and may affect downstream applications (Duwairi et al., 2007).

A.2 Intrinsic Evaluation

To complement the downstream evaluation, we conducted an intrinsic evaluation of the proposed Tunisian stemming resource. From the final manually annotated corpus, a held-out set of 800 Tunisian words was selected for intrinsic evaluation.

The hybrid stemming approach described in Section 3.3 was evaluated by comparing its automatically predicted stems against the manually annotated reference stems. The evaluation was performed separately for light and aggressive stemming using exact-match accuracy, where a prediction was considered correct only when the predicted stem exactly matched the corresponding reference stem.

The hybrid stemmer achieved an accuracy of 92.8% for light stemming and 86.1% for aggressive stemming. These results indicate that light stemming is more consistent and easier to recover than aggressive stemming. The lower performance of aggressive stemming reflects the greater difficulty of identifying the underlying lexical form in the presence of Tunisian morphological and orthographic variation.

The analysis of failures indicates that most errors fall into two main categories: orthographic variation and ambiguous dialectal forms.

Orthographic variants. Tunisian words may occur with different orthographic realizations while referring to the same lexical item. Such variation can make exact stem matching difficult even when the predicted stem is linguistically plausible.

Ambiguous dialectal forms. Some words admit different stem interpretations depending on dialectal or morphological analysis. This issue is particularly relevant for aggressive stemming, which attempts to recover a deeper lexical representation.

Overall, the intrinsic evaluation confirms that the proposed resource provides reliable stemming annotations, with particularly strong performance for light stemming.

A.3 Additional Analyses

A.3.1 Fuzzy Matching Contribution

We conducted an ablation analysis to quantify the contribution of fuzzy matching beyond exact dictionary lookup. Table 6 reports the proportion of tokens processed by each matching strategy for the two downstream tasks.

Table 6: Distribution of matching strategies (%).

Task	Dictionary	Fuzzy	Unchanged
Domain classification	81.7	12.9	5.4
Sentiment analysis	84.2	10.6	5.2

Fuzzy matching successfully recovers 12.9% and 10.6% of tokens for domain classification and sentiment analysis, respectively, that are not directly covered by the stemming lexicon. As shown in Table 7, the additional coverage improves SVM accuracy for domain classification by 1.8 percentage points compared with dictionary lookup alone.

In contrast, the contribution of fuzzy matching to sentiment analysis is limited. These results indicate that fuzzy matching provides complementary lexical coverage, while its downstream benefit remains task-dependent.

Table 7: Ablation results comparing dictionary lookup and the hybrid strategy for SVM on domain classification.

Preprocessing strategy	Accuracy
Dictionary lookup only	0.4608
Dictionary + fuzzy matching	0.4788
Absolute improvement	+1.8 points

A.3.2 Statistical Significance

We assessed the statistical significance of the differences between preprocessing configurations using McNemar’s test for pairwise comparisons. The improvement of LightStem over No Stemming for SVM on domain classification is statistically significant ($p < 0.01$). Conversely, the differences between preprocessing configurations for sentiment analysis are not statistically significant ($p > 0.05$).

These results provide statistical support for the benefit of light stemming in domain classification, while confirming that its effect on sentiment analysis is not statistically significant.